

Task 2 & 3: Exploratory Data Analysis & Data Visualization

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Task 2: Exploratory Data Analysis (EDA)

Data Loading and Structure

The dataset was loaded using Pandas. Pandas was used to examine columns, data types, and non-null counts. And functions were used to get summary statistics for numerical features. The dataset contains passenger information such as Passenger, Survived, Pclass, Name, Sex, Age, SibSp, Parch, Ticket, Fare, Cabin, and Embarked. The target variable is Survived.

Trends, Patterns, and Anomalies Identified

The bar chart of the 'Survived' column revealed a class imbalance: 69% of passengers did not survive, while 31% survived. This is a key pattern in the data and indicates that the dataset is imbalanced between the two classes.

Hypothesis Testing and Validation

Hypothesis: The dataset is complete with no missing values.

Validation: The function was applied to all columns, and a heatmap was plotted using Seaborn to visualize missing values. The result confirmed zero missing values across the dataset, validating the hypothesis.

Data Issues Detected

No missing values, outliers, or anomalies were detected in the initial inspection. The dataset is clean and ready for further analysis and visualization.

Task 3: Data Visualization

Tools Used

Matplotlib and Seaborn in Python were used to create visualizations that enhance understanding and reveal insights clearly.

Visuals Created

Survival Count Distribution

The bar chart shows the count of passengers who survived versus those who did not survive. It clearly illustrates the class imbalance in the dataset.

Missing Values Heatmap

The heatmap confirms that there are no missing values in the dataset. All cells are filled, indicating a complete dataset.

Insights and Data Story

The visualizations support the findings from the EDA. The bar chart highlights the imbalance in survival rates, which is important for any future classification model. The heatmap validates that no data cleaning for missing values is required, allowing the analysis to proceed directly to modelling or deeper exploration.

Conclusion

The Titanic dataset is complete and clean, with a clear class imbalance in the target variable. The visualizations created effectively communicate these findings and support decision-making for further analysis. This project demonstrates the ability to transform raw data into clear visual formats and craft compelling data stories.